

Centre Number	Candidate Number	Name
---------------	------------------	------

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

PHYSICS

9702/02

Paper 2

May/June 2004

1 hour

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The number of marks is given in brackets [] at the end of each question or part question.
You may lose marks if you do not show your working or if you do not use appropriate units.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
Total	

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

This document consists of **17** printed pages and **3** blank pages.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas,	$W = p\Delta V$
gravitational potential,	$\phi = -\frac{Gm}{r}$
simple harmonic motion,	$a = -\omega^2 x$
velocity of particle in s.h.m.,	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
resistors in series,	$R = R_1 + R_2 + \dots$
resistors in parallel,	$1/R = 1/R_1 + 1/R_2 + \dots$
electric potential,	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series,	$1/C = 1/C_1 + 1/C_2 + \dots$
capacitors in parallel,	$C = C_1 + C_2 + \dots$
energy of charged capacitor,	$W = \frac{1}{2}QV$
alternating current/voltage,	$x = x_0 \sin \omega t$
hydrostatic pressure,	$p = \rho gh$
pressure of an ideal gas,	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
radioactive decay,	$x = x_0 \exp(-\lambda t)$
decay constant,	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
critical density of matter in the Universe,	$\rho_0 = \frac{3H_0^2}{8\pi G}$
equation of continuity,	$Av = \text{constant}$
Bernoulli equation (simplified),	$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$
Stokes' law,	$F = Ar\eta v$
Reynolds' number,	$R_e = \frac{\rho v r}{\eta}$
drag force in turbulent flow,	$F = Br^2 \rho v^2$

Answer **all** the questions in the spaces provided.

- 1 (a) State the difference between a scalar quantity and a vector quantity.

scalar:

.....

vector:

..... [2]

- (b) Two forces of magnitude 6.0 N and 8.0 N act at a point P. Both forces act away from point P and the angle between them is 40° .
Fig. 1.1 shows two lines at an angle of 40° to one another.

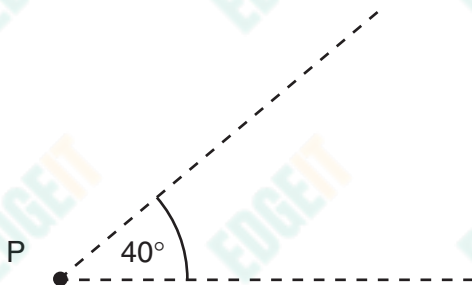


Fig. 1.1

On Fig. 1.1, draw a vector diagram to determine the magnitude of the resultant of the two forces.

magnitude of resultant = N [4]

- 2 Fig. 2.1 shows the variation with distance x along a wave of its displacement d at a particular time.

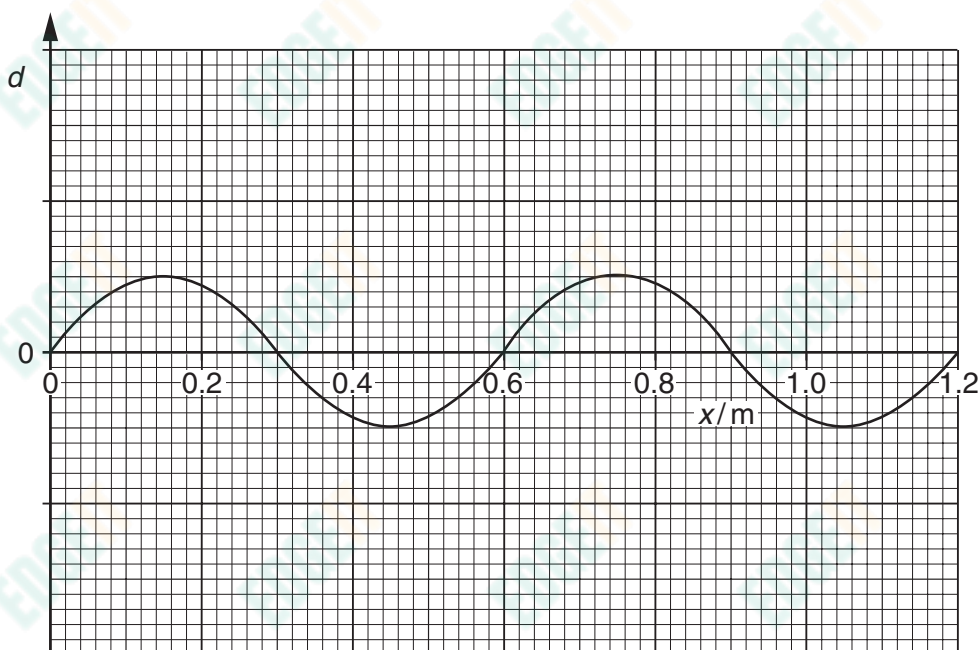


Fig. 2.1

The wave is a progressive wave having a speed of 330 m s^{-1} .

- (a) (i) Use Fig. 2.1 to determine the wavelength of the wave.

wavelength = m

- (ii) Hence calculate the frequency of the wave.

frequency = Hz
[3]

- (b) A second wave has the same frequency and speed as the wave shown in Fig. 2.1 but has double the intensity. The phase difference between the two waves is 180° .

On the axes of Fig. 2.1, sketch a graph to show the variation with distance x of the displacement d of this second wave.
[2]

- 3 A student has been asked to determine the linear acceleration of a toy car as it moves down a slope. He sets up the apparatus as shown in Fig. 3.1.

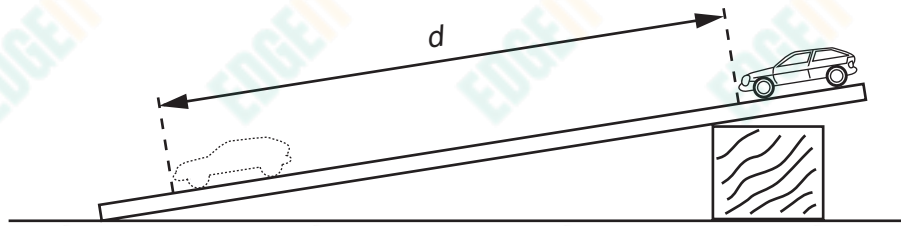


Fig. 3.1

The time t to move from rest through a distance d is found for different values of d . A graph of d (y-axis) is plotted against t^2 (x-axis) as shown in Fig. 3.2.

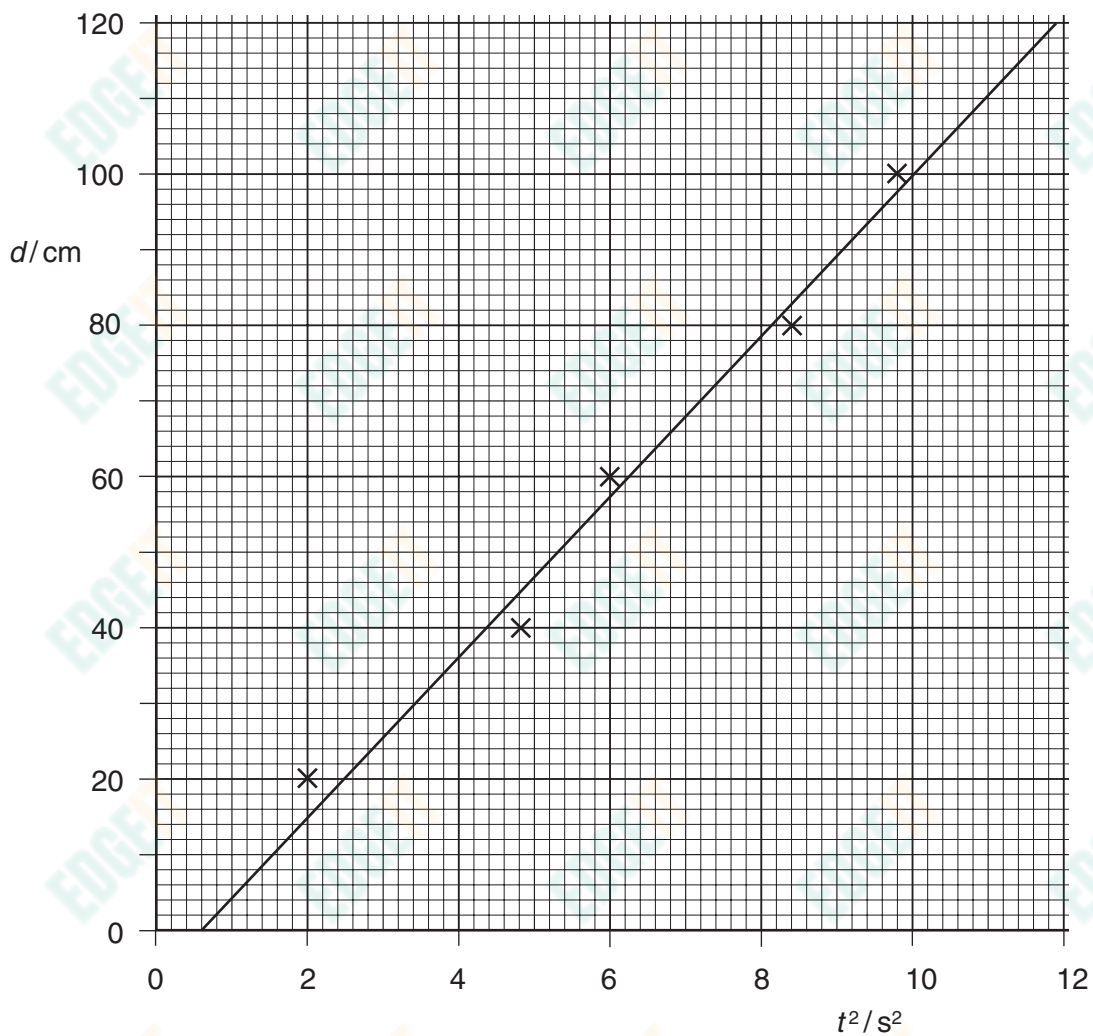


Fig. 3.2

- (a) Theory suggests that the graph is a straight line through the origin.
Name the feature on Fig. 3.2 that indicates the presence of

(i) random error,

.....

(ii) systematic error.

.....

[2]

- (b) (i) Determine the gradient of the line of the graph in Fig. 3.2.

gradient = [2]

- (ii) Use your answer to (i) to calculate the acceleration of the toy down the slope.
Explain your working.

acceleration = ms^{-2} [3]

- 4 A ball has mass m . It is dropped onto a horizontal plate as shown in Fig. 4.1.

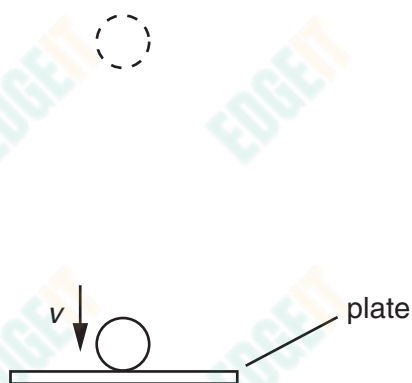


Fig. 4.1

Just as the ball makes contact with the plate, it has velocity v , momentum p and kinetic energy E_k .

- (a) (i) Write down an expression for momentum p in terms of m and v .

.....

- (ii) Hence show that the kinetic energy is given by the expression

$$E_k = \frac{p^2}{2m}.$$

[3]

- (b) Just before impact with the plate, the ball of mass 35 g has speed 4.5 m s^{-1} . It bounces from the plate so that its speed immediately after losing contact with the plate is 3.5 m s^{-1} . The ball is in contact with the plate for 0.14 s.

Calculate, for the time that the ball is in contact with the plate,

- (i) the average force, in addition to the weight of the ball, that the plate exerts on the ball,

magnitude of force = N

direction of force =
[4]

- (ii) the loss in kinetic energy of the ball.

loss = J [2]

- (c) State and explain whether linear momentum is conserved during the bounce.

.....
.....
.....
..... [3]

- 5 Two forces, each of magnitude F , form a couple acting on the edge of a disc of radius r , as shown in Fig. 5.1.

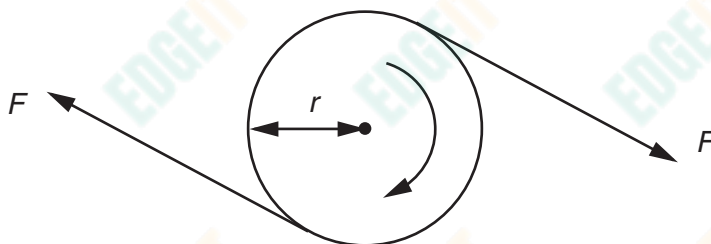


Fig. 5.1

- (a) The disc is made to complete n revolutions about an axis through its centre, normal to the plane of the disc. Write down an expression for

- (i) the distance moved by a point on the circumference of the disc,

distance =

- (ii) the work done by one of the two forces.

work done = [2]

- (b) Using your answer to (a), show that the work W done by a couple producing a torque T when it turns through n revolutions is given by

$$W = 2\pi nT. \quad [2]$$

- (c) A car engine produces a torque of 470 N m at 2400 revolutions per minute. Calculate the output power of the engine.

power = W [2]

- 6 Fig. 6.1 shows wavefronts incident on, and emerging from, a double slit arrangement.

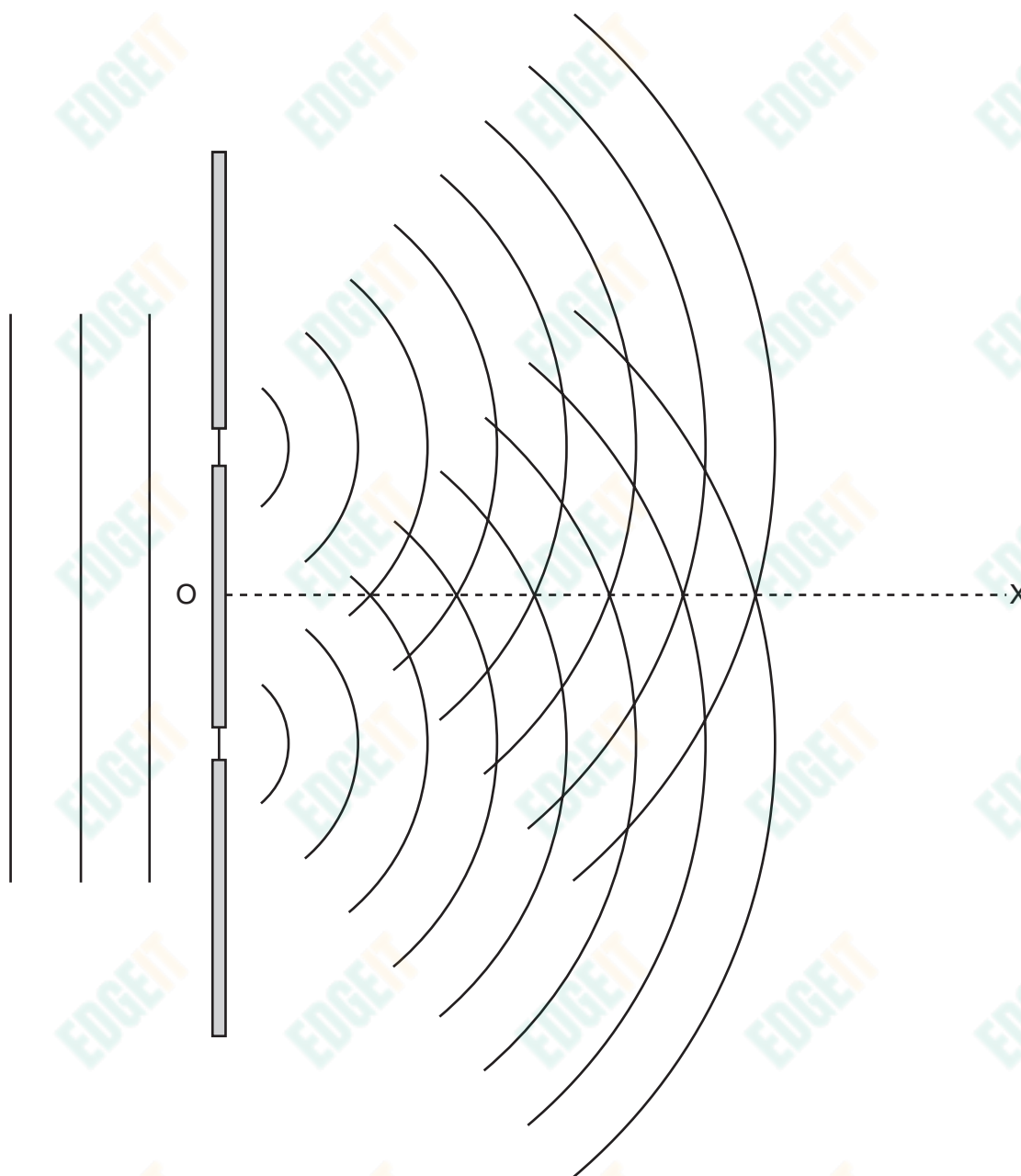


Fig. 6.1

The wavefronts represent successive crests of the wave. The line OX shows one direction along which constructive interference may be observed.

- (a)** State the principle of superposition.

.....

.....

..... [3]

(b) On Fig. 6.1, draw lines to show

- (i) a second direction along which constructive interference may be observed (label this line CC),
- (ii) a direction along which destructive interference may be observed (label this line DD).

[2]

(c) Light of wavelength 650 nm is incident normally on a double slit arrangement. The interference fringes formed are viewed on a screen placed parallel to and 1.2 m from the plane of the double slit, as shown in Fig. 6.2.

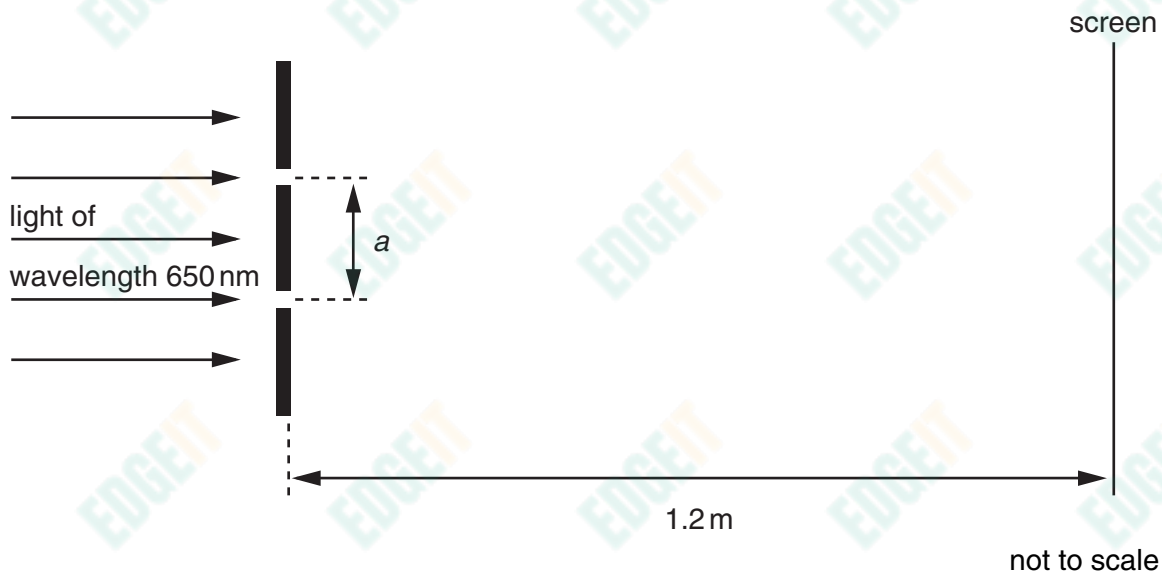


Fig. 6.2

The fringe separation is 0.70 mm.

- (i) Calculate the separation a of the slits.

separation = m [3]

(ii) The width of both slits is increased without changing their separation a . State the effect, if any, that this change has on

1. the separation of the fringes,

.....

2. the brightness of the light fringes,

.....

3. the brightness of the dark fringes.

.....

[3]

- 7 A household electric lamp is rated as 240 V, 60 W. The filament of the lamp is made from tungsten and is a wire of constant radius $6.0 \times 10^{-6} \text{ m}$. The resistivity of tungsten at the normal operating temperature of the lamp is $7.9 \times 10^{-7} \Omega \text{ m}$.

(a) For the lamp at its normal operating temperature,

(i) calculate the current in the lamp,

current = A

(ii) show that the resistance of the filament is 960Ω .

[3]

(b) Calculate the length of the filament.

length = m [3]

(c) Comment on your answer to (b).

.....
..... [1]

- 8 A thermistor has resistance $3900\ \Omega$ at 0°C and resistance $1250\ \Omega$ at 30°C . The thermistor is connected into the circuit of Fig. 8.1 in order to monitor temperature changes.

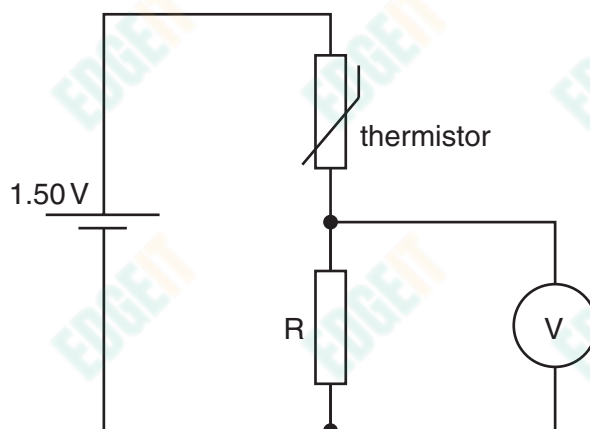


Fig. 8.1

The battery of e.m.f. $1.50\ \text{V}$ has negligible internal resistance and the voltmeter has infinite resistance.

- (a) The voltmeter is to read $1.00\ \text{V}$ at 0°C . Show that the resistance of resistor R is $7800\ \Omega$.

[2]

- (b) The temperature of the thermistor is increased to 30°C . Determine the reading on the voltmeter.

reading = V [2]

- (c) The voltmeter in Fig. 8.1 is replaced with one having a resistance of $7800\ \Omega$. Calculate the reading on this voltmeter for the thermistor at a temperature of $0\ ^\circ\text{C}$.

reading = V [2]

Every reasonable effort has been made to trace all copyright holders where the publishers (i.e. UCLES) are aware that third-party material has been reproduced. The publishers would be pleased to hear from anyone whose rights they have unwittingly infringed.

University of Cambridge International Examinations is part of the University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.